# Zixing JIANG

Robotics and Artificial Intelligence Laboratory (RAIL)
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## EDUCATION

B.Eng. Electronic Information Engineering (Stream: Computer Engineering)
The Chinese University of Hong Kong, Shenzhen (CUHK-Shenzhen)
Shenzhen, Guangdong province, China
Sept. 2019 – May 2023

- Completed with Honours, First Class.
- Final Year Project: *Control of the Multi-Joint Manipulator for Grasping on Water Surface.*
- Core curriculum: *Robotics and Intelligent Systems, Automatic Control Theory, Programming for Robotics, Image Processing and Computer Vision, Reinforcement Learning, Artificial Intelligence, Parallel Programming, Signals and Systems*, etc.

## RESEARCH INTERESTS

Image- and force-guided robotic systems, Medical robotics, Bio-inspired robotics

## RESEARCH EXPERIENCE

2023– Research Assistant
Surgical Robotics Group, RAIL, CUHK-Shenzhen
Supervisor: Prof. Zhenglong Sun
Mar. 2023 – Present

- Ongoing. Leading a autonomous robotic ultrasound project, forcing on thyroid and breast.

2020–23 Research Assistant
Advanced Marine Robotics Group, RAIL, CUHK-Shenzhen
Supervisor: Prof. Huihuan Alex Qian
Sept. 2020 – Feb. 2023

- Assisted in research on manipulator-assisted drone landing and floating-base manipulator.
- Responsibilities included building robot prototypes (mechanical design, electronics design, and robot programming), conducting experiments and drafting research papers.

2021–22 Undergraduate Research Award (URA) Awardee
School of Science and Engineering (SSE), CUHK-Shenzhen
Advisors: Prof. Xiaoqiang Ji, Prof. Huihuan Qian, Prof. Zhenglong Sun
Nov. 2021 – Oct. 2022

- Built a kingfisher-inspired flapping-wing robot prototype, financially supported by CUHK-Shenzhen URA committee.

## PUBLICATIONS[1]

### Under Review

2023 Ruoyu Xu, **Zixing Jiang**, Beibei Liu, Yuquan Wang and Huihuan Qian*, "Confidence-Aware Object Capture for a Manipulator Subject to Floating-Base Disturbances", Submitted to *IEEE Transactions on Robotics (T-RO)*. 2023

### Conference Papers

2023 Yiyang Jiang, Ruoyu Xu, **Zixing Jiang**, and Huihuan Qian*, "Design, Modeling and Control of A Novel USV-Manipulator System", *2023 IEEE International Conference on Real-time Computing and Robotics (RCAR)*, IEEE, 2023

2022 Chongfeng Liu, **Zixing Jiang**, Ruoyu Xu, Xiaoqiang Ji, Lianxin Zhang, and Huihuan Qian*, "Design and Optimization of a Magnetic Catch System for Precise UAV Landing on Disturbed Aquatic Surface Platform", *2022 IEEE International Conference on Robotics and Automation (ICRA)*, IEEE, 2022, pp. 1162-1168

### Patents

2022 Chongfeng Liu, Zhongzhong Cao, **Zixing Jiang**, Rouyu Xu, Xiaoqiang Ji, and Huihuan Qian, "Landing System, Landing Method and Storage Medium for Unmanned Aerial Vehicle", Chinese patent CN115167522A (initiative for examination as to substance)

2022 **Zixing Jiang**, Xiaoqiang Ji, Chongfeng Liu, and Huihuan Qian, "A Quadcopter Flapping-Wing Micro Aerial Vehicle and its Flight Method", Chinese patent CN114889821A

2022 Xiaoqiang Ji, Zefang Song, **Zixing Jiang**, and Huihuan Qian, "A Flapping-wing Configuration and Flapping-Wing Micro Aerial Vehicle on Water Surface", Chinese patent CN217320745U

2022 Zefang Song, Xiaoqiang Ji, **Zixing Jiang**, and Huihuan Qian, "A Dual Crank-based Flapping-wing Configuration and Flapping-Wing Micro Aerial Vehicle on Water Surface", Chinese patent CN217320744U

## SELECTED PROJECTS

### Final Year Project (FYP)

2022 Control of the Multi-Joint Manipulator for Grasping on Water Surface
Advanced Marine Robotics Group, RAIL, CUHK-Shenzhen
Supervisor: Prof. Huihuan Alex Qian
Sept. 2022 – Dec. 2022

- Proposed a predictive control framework that makes manipulators capable of grasping objects on disturbed platforms. The control framework consists of minimum-volume enclosing ellipsoid fitting, wavelet regression based motion prediction, cubic polynomial trajectory generation and Levenberg-Marquardt inverse kinematics.
- Implemented the proposed control framework with MATLAB and verified its feasibility in Simscape simulation environment.
- Completed with Distinction, SSE 2019 cohort, CUHK-Shenzhen.

[1]Notation: # equal contribution, * corresponding author

## Research Projects

2023– Autonomous Robotic Ultrasound for Thyroid and Breast
Surgical Robotics Group, RAIL, CUHK-Shenzhen
Supervisor: Prof. Zhenglong Sun
Mar. 2023 – Present

- Ongoing. This project aims to develop an autonomous robotic ultrasound system for thyroid and breast, with the commercial Universal Robots UR5e platform.
- Developing visual servoing and hybrid force/motion control strategies based on real-time ultrasound image and end-effector 6-axis wrench feedback, respectively.

2023 Confidence-Aware Object Capture for a Floating-Base Manipulator
Advanced Marine Robotics Group, RAIL, CUHK-Shenzhen
Supervisor: Prof. Huihuan Alex Qian
Jan. 2023 – May 2023

- Worked closely with a doctorial postgraduate to improve the predictive control framework proposed in my FYP by introducing confidence analysis.
- Assisted in implementing and testing the improved control framework on a real robot. Achieved a 0.2 second response time and 80% capture accuracy.
- This work was summarized as a journal article and submitted to IEEE T-RO.

2021–22 Bio-Inspired Aquatic-Aerial Hybrid Locomotion Robot
Undergraduate Research Award & RAIL, CUHK-Shenzhen
Advisors: Prof. Xiaoqiang Ji, Prof. Huihuan Alex Qian, Prof. Zhenglong Sun
Nov. 2021 – Oct. 2022

- Inspired by the kingfisher, built a flapping-wing driven robot prototype that can perform both aquatic and aerial locomotion.
- Some achievements of the project were granted Chinese patents.

2020–21 Precise Landing of Unmanned Aerial Vehicle (UAV) on Disturbed Aquatic Surface Platforms
Advanced Marine Robotics Group, RAIL, CUHK-Shenzhen
Supervisor: Prof. Huihuan Alex Qian
Sept. 2020 – Sept. 2021

- Proposed a tethered landing method with a doctorial postgraduate to reduce the difficulty of landing UAVs precisely on disturbed surface platforms in turbulent water.
- Assisted in designing and fabricating a magnetic catch system to implement the proposed tethered landing method.
- This work was summarized as a conference paper and presented to IEEE ICRA 2022.

## Course Projects

2023 Adaptive Dynamic Stabilization of a Self-Balancing Scooter Using Epsilon-Greedy Q-Learning
4-Person Team Project, Team Leader & Major Contribution
CSC3180: Fundamentals of Artificial Intelligence, 2023 Spring, CUHK-Shenzhen
Course Instructor: Prof. David Zhang Dapeng

- Led three classmates to develop a reinforcement learning-based algorithm that enables self-balancing vehicles to adapt to loads with different centers of gravity and remain balance.
- Validated the proposed algorithm on the OpenAI Gym simulation environment as well as on a real robot.

2022 Manipulator Assembly and Kinematics Control
2-Person Team Project, Team Leader & Dominant Contribution
CIE6107: Robotics and Intelligent Systems, 2022 Fall, CUHK-Shenzhen
Course Instructor: Prof. Huihuan Alex Qian

- Assembled a 5-degree-of-freedom serial manipulator from parts and implemented its forward and inverse kinematics with Denavit-Hartenberg method.
- Programmed the manipulator to perform tasks like pick-and-throw with MATLAB and Arduino.

2022 Depth Estimation with Single-Camera Motion Stereo in Visual-Aid Odometry
2-Person Team Project, Team Leader & Dominant Contribution
ECE4513: Image Processing and Computer Vision, 2022 Fall, CUHK-Shenzhen
Course Instructor: Prof. Zhen Li

- Derived a depth estimation algorithm of monocular camera through camera intrinsic. With the prior knowledge of a moving object's speed, the algorithm can estimate the depth of the object relative to the camera through the parallax of the object in two consecutive frames.
- Applied the proposed depth estimation algorithm to robot odometry. A MATLAB simulation showed that the visual information obtained by this algorithm can reduce the cumulative error of the robot IMU.

2022 Intelligent Sorting with Eye-in-Hand Manipulator
Individual Project
ECE4310: Programming for Robotics, 2022 Spring, CUHK-Shenzhen
Course Instructor: Prof. Tim Lun Lam

- Programmed a 6-degree-of-freedom serial manipulator with eye-in-hand camera configuration to perform an intelligent sorting task: automatically places the different colored pieces into the corresponding colored trash bins.
- Used the Robot Operating System (ROS) MoveIt! package for robot kinematics. Utilized linear regression for hand-eye calibration. Implemented color recognition with the OpenCV library and HSV masks.

2022 Navigation for Service Robot
Individual Project
ECE4310: Programming for Robotics, 2022 Spring, CUHK-Shenzhen
Course Instructor: Prof. Tim Lun Lam

- Programmed a service robot to perform an auto-delivery task in ROS Gazebo simulation environment: traverse all delivery sites; each site is distinguished with an Artificial Reality (AR) tag, which indicates the position of the following site relative to the current site.
- Used ROS move_base package for robot path planning. Used LiDAR to construct local map for obstacle avoidance. Used ROS ar_track_alvar package for AR tag recognition.

## EXTRACURRICULAR ACTIVITIES

2022 Hair Donor of Qingsi Xingdong (青丝行动)
Dec. 2022

- Qingsi Xingdong is a non-profit charity organization founded by Chinese university students that accepts donated hair, makes wigs, and distributes them free of charge to cancer

patients undergoing chemotherapy.

2019-22 Chair of Student Robotics Association (SRA), RAIL, CUHK-Shenzhen
Sept. 2020 – Feb. 2022

- Student Robotics Association (SRA) was held by Prof. Huihuan Alex Qian in RAIL, CUHK-Shenzhen. Aiming to bridge CUHK-Shenzhen undergraduates and RAIL, it offers robotic training and hands-on projects for incoming undergraduate students to get prepared for research activities before joining the lab.
- I coordinated fellow junior SRA members to recruit new members, holding RAIL briefings, deliver weekly tutorials (PID control, microcontroller programming, and robot kinematics) and mentor new SRA members to complete hands-on projects (unmanned sailboats and unmanned McNamee wheeled vehicles). We successfully recruited more than 30 members and over a half of them managed to start working at the lab.

## COMPETITIONS

2021 DJI RoboMatser University AI Challenge (RMUA)
9-Person CUHK-Shenzhen Varsity

- Served as the leader of a 3-person robot perception team in the CUHK-Shenzhen RMUA varsity. Implemented a vehicle visual tracking system based on YOLO and KCF algorithms with the team, which can achieve about 80% tracking accuracy.

2020 RoboCom World Robotics Developer Competition, Spark ROS Robotics Competition
5-Person Team

- Responsible for robot hardware debugging and robot programming.
- Achieved 2nd Class Award of the team in Semi-Autonomous Competition and 3rd Class Award of the team in Autonomous Competition, China Guangdong Division.

## HONOURS & AWARDS

2023 Undergraduate First Class Honours Degree, CUHK-Shenzhen

2023 Final Year Project Distinction, CUHK-Shenzhen

2021–22 17th–19th Round Undergraduate Research Award, CUHK-Shenzhen

2020 Spark ROS Robotics Competition, RoboCom 2020
- 2nd Class Award of the team in Semi-Autonomous Competition, Guangdong Division
- 3rd Class Award of the team in Autonomous Competition, Guangdong Division

## SKILLS

- **Languages**: Mandarin Chinese (native speaker), English
- **Programming**: C/C++, Python, and Bash on Linux (Ubuntu, CentOS) platform
- **Robotics development toolkit**: MATLAB, ROS (1 $\gg$ 2) suite, STM32, Arduino, Solidworks
- **Algorithms**: manipulator kinematics and dynamics, motion planning and control, force control, image processing and computer vision, machine learning and reinforcement learning

• **Technical interests**: typesetting systems (LaTeX, typst), mathematical animation engine (Manim), computer graphics, multiphysics simulation, compiler (LLVM)

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